

ME 511: Manufacturing of Composites

Course Code	ME 511
Course Name	Manufacturing of Composites
L-T-P-C	3-0-0-3
Prerequisites	Basic Manufacturing Course
Intended for	B.Tech. (4th year onwards)/M.Tech./M.Tech.(R)/Ph.D.
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Composites:** Function of the Matrix and Reinforcement in Composites Matrices: Thermosets and Thermoplastic; Fiber Reinforcement (3 Hours)
- **Properties and testing composites:** Properties of Composites; Composites testing; Composites design: Laminate theory, Rule of mixtures, symmetry and balance (6 Hours)
- **Thermoset composite manufacturing processes:** Lay-up processes, spray up process; Thermoset Composite manufacturing: Fiber placement process; Thermoset Composite manufacturing: Resin transfer moulding, Vacuum assisted resin infusion microwave curing, recycling of thermoset composites, latest topics in thermoset composite manufacturing. (6 Hours)
- **Thermoplastic composite manufacturing processes:** Thermoset Composite manufacturing: Vacuum assisted resin transfer moulding; Thermoset Composite manufacturing: Compression molding process; Thermoset composites manufacturing: Filament winding, Microwave assisted Compression moulding, Additive manufacturing techniques for thermoplastic composites, latest topics in thermoplastic composite manufacturing. (6 Hours)
- **Metal and Ceramic Matrix Composites:** Metal Matrix Composites: Metal matrix and reinforcement; Manufacturing processes for Metal Matrix Composites: Dispersion hardened and particle composite; Manufacturing processes for Metal matrix composites: Layer composites and infiltration method; Ceramic Matrix Composite manufacturing, latest topics in metal and ceramic matrix composite manufacturing. (7 Hours)
- **Secondary Manufacturing Techniques for Composites:** Joining techniques: Hot plate welding; Ultrasonic joining; Adhesive binding, composite repair techniques. Machining techniques for composites Machining Techniques for composite, Laser beam machining, electric discharge machining, ultrasonic machining, water jet machining, conventional drilling, milling and turning operations. (7 Hours)
- **Process modelling in Composite Manufacturing:** Transport equations for composite processing, constitutive laws and their characterization, Resin viscosity, Reaction kinetics, crystallization kinetics, model simplification and solution, application of numerical model in short fiber composites, thermoplastic composites, thermoset composites. (7 Hours)

Text books: (Relevant and Latest, Only 2)

1. Strong AB, **Fundamentals of composites manufacturing: materials, methods and applications**, Society of manufacturing engineers, 2008.
2. Mallick PK, **Fiber-reinforced composites: materials, manufacturing, and design**, CRC press, 2007.

References: (No limit on numbers, relevant standard format can be followed, the formats should be similar)

1. ASM International Handbook Committee, **Composites: Volume 21 of ASM Handbook**.